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DTE Energy

10 CFR 50.73

December 20, 2016
NRC-16-0046

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2016-009

Pursuant to 10 CFR 50.73(a)(2)(i)(B) and 50.73(a)(2)(v)(B) and (D), DTE Electric Company (DTE) is submitting LER No. 2016-009, Emergency Diesel Generator Inoperable Due to Open Circuit on Loss of Power Instrumentation.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Scott A. Maglio, Manager – Nuclear Licensing, at (734) 586-5076.

Sincerely,

Keith J. Polson
Site Vice President

Enclosure: Licensee Event Report No. 2016-009

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov)

Enclosure to
NRC-16-0046

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2016-009

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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4. TITLE

Emergency Diesel Generator Inoperable Due to Open Circuit on Loss of Power Instrumentation

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME: N/A	DOCKET NUMBER: 05000
04	24	2016	2016	009	00	12	20	2016	FACILITY NAME: N/A	DOCKET NUMBER: 05000

9. OPERATING MODE: 1

10. POWER LEVEL: 100

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☑ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER (Include Area Code)
Fermi 2 / Scott A. Maglio – Manager, Nuclear Licensing	(734) 586-5076

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
X	EK	XPT	G080	Y	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☑ NO

15. EXPECTED SUBMISSION DATE MONTH DAY YEAR

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

At 1644 EDT on April 24, 2016, while operating at 100 percent Reactor Thermal Power, the potential transformer (PT) primary fuses on Bus 64C were found blown, resulting in half of the Loss of Power (LOP) instrumentation relays tripping on Bus 64C. Initially, the LOP instrumentation was incorrectly considered OPERABLE; however, later reviews of the condition determined that the LOP instrumentation was inoperable. Later reviews also concluded that the Division 1 Alternating Current (AC) Source was inoperable from 1644 to 1920 on April 24, 2016, and the Division 1 Emergency Diesel Generator (EDG) 12 was inoperable starting at 1644 on April 24, 2016, until it was restored at 1818 on May 8, 2016, during a Forced Outage following replacement of the blown PT fuses. On six occasions from April 26-30, 2016, Division 2 EDGs 13 and/or 14 were inoperable for planned surveillances. Therefore, this is an event or condition that could have prevented the fulfillment of a safety function to remove residual heat and mitigate the consequences of an accident. Furthermore, failure to enter applicable Limiting Conditions for Operation resulted in a condition prohibited by Technical Specifications. The cause of the event was concluded to be a low energy transient event resulting in the blown PT primary fuses. The incorrect initial operability determination was caused by Operators being unfamiliar with this equipment condition. Operators completed a Department Event Free Day Reset and Event Review and will receive additional training on this condition.

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NARRATIVE

INITIAL PLANT CONDITIONS

Mode – 1
Reactor Power – 100 percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At 1644 EDT on April 24, 2016, while operating in Mode 1 at 100 percent Reactor Thermal Power, Operators received alarm 9D8, Transformer [[XFMR]] 64 (SS64) Trouble. An initial investigation revealed that the Potential Transformer (PT) [[XPT]] for Buses [[BU]] 64A and 64C was not providing an output voltage signal to the relaying and instrumentation downstream of the PTs. Half of the loss of voltage (LOV) relays [[27]] and half of the degraded voltage relays (DVRs) for the 4.16 kiloVolt (kV) Emergency Bus 64C undervoltage were tripped while the other half of both of those relays were not tripped. Therefore, the LOP instrumentation trip logic was in a half-tripped state. The LOV relays and DVRs together constitute the Technical Specification (TS) 3.3.8.1 Loss of Power (LOP) instrumentation to detect and respond to a loss of offsite power (LOOP) event.

Licensed Operators initially incorrectly considered SS64 OPERABLE based on the proper voltage being maintained on Engineered Safety Feature (ESF) Buses 64B and 64C. Initially, the Division 1 Bus 64C LOP instrumentation was also incorrectly considered OPERABLE by Licensed Operators because the relays failed in their fail safe condition and the logic was capable of performing the safety function. The failure resulted in the LOP instrumentation going from a one out of two, taken twice, logic to a one out of two logic, which would have properly detected and responded to a loss of offsite power or degraded voltage event.

At 1920 EDT on April 24, 2016, Operators restored the SS64 output voltage to above 121.3 meter Volts (V) by manual operation of the Load Tap Changer (LTC) [[TTC]] as required by System Operating Procedure 23.320, "Balance of Plant Auxiliary Electrical Distribution System" (Revision 63). Restoration of the voltage to 121.3 meter V ensured that adequate voltage was available to reset the LOV relays and DVRs during a LOCA coincident with a degraded grid.

Division 2 offsite power and Combustion Turbine Generator (CTG) [[88]] 11-1 were OPERABLE at all times during the event.

Although the equipment was initially determined to be OPERABLE, repairs were still required. Fermi 2 entered Mode 3 at 2301 EDT on May 3, 2016, and subsequently, entered Mode 4 at 1525 EDT on May 4, 2016, for Forced Outage (FO) 16-01 to trouble-shoot and repair the LOP instrumentation.

Subsequent analysis of this event by Engineering determined that:

A. Starting at 1644 on April 24, 2016, with the LTC automatic function disabled and voltage below 121.3 meter V, per Design Calculation (DC) 6447 Volume I, "Auxiliary Power System Analysis," (Revision C) the DVRs may not have reset upon a Loss of Coolant Accident (LOCA) auto-start of ESF equipment in order to stay connected to the preferred offsite power source. Manual operation of the SS64 LTC by Operations boosted the output voltage of SS64 and ensured adequate voltage to reset the DVRs at 1920 on April 24, 2016.

NARRATIVE

TS Limiting Condition for Operation (LCO) 3.3.8.1 requires the LOP instrumentation to have four functional channels [[CHA]] for each Division. Resetting of the LOV relays and DVRs is not an explicit TS 3.3.8.1 Surveillance Requirement (SR) although the reset of the LOV relays and DVRs is credited in the electrical analysis. With both the LOV relays and DVRs (connected across phase-to-neutral of the line side PTs) remaining tripped, there was no way to reset these tripped relays in situ to return to initial conditions or to apply the existing test procedure to test the LOP instrumentation channels for compliance with TS SRs 3.3.8.1.1 and 3.3.8.1.3. Therefore, the LOP instrumentation was inoperable from 1644 on April 24, 2016, until 1818 on May 8, 2016.

B. Between 1644 and 1920 on April 24, 2016, Division 1 Alternating Current (AC) Source lost its functionality due to the inability of SS64 to deliver three phase AC power at the required voltage range stemming from its loss of voltage reference required for LTC automatic operation. SS64 output voltage was restored at 1920 to above 121.3 meter V by manual operation of the LTC in accordance with System Operating Procedure 23.320. Therefore, Division 1 AC Source was inoperable from 1644 to 1920 on April 24, 2016.

C. At 1644 on April 24, 2016, Emergency Diesel Generator (EDG) [[DG]] 12 lost its voltage reference required for synchronization across Breaker [[BKR]] C6 upon restoration of offsite power source. EDG 12 would not have been able to pass TS SR 3.8.1.15, which requires each EDG to synchronize with offsite power while loaded with emergency loads upon a simulated restoration of offsite power, transfer loads to offsite power, and return to standby status. The loss of synchronization capability lasted until the primary fuses for the ESF Bus 64C line side PTs were replaced on May 7, 2016, during FO 16-01. The loss of voltage reference did not affect the ability of EDG 12 to start and load in an accident scenario, but EDG 12 could not be manually synchronized with offsite power to allow loads to be transferred back to offsite power. Therefore, EDG 12 was inoperable from 1644 on April 24, 2016, until 1818 on May 8, 2016, when it was returned to service.

Based on the Engineering analysis, multiple LCOs should have been entered by Operations based on inoperability of plant equipment as a result of this event. Because these LCOs were not entered, a number of LCO Conditions were not met. Specifically:

1. Due to the Division 1 AC Source being inoperable, LCO 3.8.1, Condition D for one inoperable offsite circuit should have been entered from 1644 to 1920 on April 24, 2016. Required Action D.1, which requires performing TS SR 3.8.1.1 for the OPERABLE offsite circuit, has a Completion Time of 1 hour, which was not met. TS SR 3.8.1.1 verifies correct breaker alignment and indicated power availability for each offsite circuit. Although TS SR 3.8.1.1 was not performed, Division 2 offsite power was available. All other Required Actions for this Condition have Completion Times that are longer than the duration of this condition.

2. Due to the LOP instrumentation being inoperable, LCO 3.3.8.1 Condition A for one or more buses with one or more channels inoperable should have been entered at 1644 on April 24, 2016. LCO 3.3.8.1 Condition A Required Action A.1 requires that the channel be restored to OPERABLE status within 72 hours. The LOP channels were not restored in 72 hours and therefore, LCO 3.3.8.1 Condition B should have been entered at 1644 on April 27, 2016. LCO 3.3.8.1 Condition B Required Action B.1 requires that the associated EDG be declared inoperable immediately when Condition A could not be met. Therefore, both of these Required Action Completion Times were not met.

NARRATIVE

3. Due to EDG 12 being inoperable starting at 1644 on April 24, 2016, LCO 3.8.1 Condition A for one EDG inoperable should have been entered until Mode 4 was entered at 1525 on May 4, 2016. LCO 3.8.1 Condition A Required Action A.1 requires that TS SR 3.8.1.1 be performed for operable offsite power circuit(s) within one hour and once per 8 hours thereafter. Although TS SR 3.8.1.1 was not performed at the required times, Division 2 offsite power was available throughout this event. LCO 3.8.1 Condition A Required Action A.2 states that required feature(s), supported by the inoperable EDGs, must be declared inoperable within 4 hours when the redundant required feature(s) are inoperable. This Required Action was not performed and during the timeframe EDG 12 was inoperable work was performed on required redundant features in Division 2. Some of this work exceeded four hours; therefore, this Required Action was not met. LCO 3.8.1 Condition A Required Action A.3 requires verifying the status of CTG 11-1 once per 8 hours, and Required Action A.5 requires restoration of CTG 11-1 within 72 hours of discovery of this condition concurrent with CTG 11-1 not being available. Required Action A.3 was not performed; however, CTG 11-1 was available throughout this time. Therefore, Required Action A.5 would not have applied. LCO 3.8.1 Condition A also has a 24 hour Completion Time to either determine operable EDGs are not inoperable due to common cause failure (Required Action A.4.1) or perform TS SR 3.8.1.2 for operable EDGs (Required Action A.4.2). Neither Required Action A.4.1 nor Required Action A.4.2 was performed at the time they were applicable; however, no EDGs were inoperable due to common cause failure as no other EDGs were affected by this condition. LCO 3.8.1 Condition A Required Action A.6 requires restoring the EDG to OPERABLE status within 14 days. As previously noted, Mode 3 was entered on May 3, 2016, and Mode 4 was entered on May 4, 2016 (approximately 10 days later). Finally, LCO 3.8.1 Condition G requires the reactor to be in Mode 3 in 12 hours if the Required Actions and associated Completion Times of Condition A are not met. Based on the shortest LCO 3.8.1 Condition A Completion Time (1 hour for Required Action A.1), the LCO 3.8.1 Condition G Completion Time was not met.

4. During the timeframe when Division 1 EDG 12 was inoperable, planned surveillances (SRs) were performed on the Division 2 EDGs (13 and 14). The following list delineates when the Division 2 EDGs were inoperable.
• EDGs 13 and 14 were both inoperable during the following timeframes:
a) 1455 to 2215 on April 27, 2016, and
b) 0033 to 0446 on April 30, 2016.
• EDG 13 was inoperable on April 26, 2016, during the following timeframes:
c) 1706 to 1806, and
d) 1957 to 2207.
• EDG 14 was inoperable on April 30, 2016, during the following timeframes:
e) 0909 to 1033, and
f) 1535 to 1611.
LCO 3.8.1 Condition C Required Action C.1 requires that if one or both EDGs in both divisions are inoperable, both EDGs in one division be restored to OPERABLE status within two hours. This LCO Condition was not entered. Timeframes a, b, and d exceeded two hours; however, timeframes c, e, and f were less than two hours. As previously noted, LCO 3.8.1 Condition G requires the reactor to be in Mode 3 in 12 hours if the Required Action and associated Completion Time of Condition C are not met. LCO 3.8.1 Condition G was not entered; however, none of the timeframes (a through f) when EDGs 13 and/or 14 were inoperable exceeded 12 hours. Therefore, this LCO Completion Time is longer than the duration of this condition.

As noted in Example 3 above, LCO 3.8.1 Condition A Required Action A.2 and an identical LCO (3.8.1 Condition B Required Action B.2 for two EDGs in one division inoperable) state that required feature(s), supported by the inoperable EDGs, must be declared inoperable within four hours when the redundant required feature(s) are inoperable. These

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Required Actions were performed during the timeframes a through f as required; however, Operators did not know at the time that EDG 12 was inoperable. If Operators had known that EDG 12 was inoperable, the SRs on Division 2 EDGs 13 and/or 14 in timeframes a through f would not have been performed.

Both EDGs in one division are required to provide AC power. The six timeframes (a through f) noted above when one or both of EDGs 13 and 14 were inoperable while EDG 12 was inoperable are reportable based on meeting the criteria of Title 10 Code of Federal Regulations (10 CFR) 50.73(a)(2)(v) (B) and (D) for an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to remove residual heat and mitigate the consequences of an accident, respectively.

Because the instances described in this LER occurred in the past and there was no loss of safety function at the time of discovery, no Event Notification was made under the corresponding requirement in 10 CFR 50.72(b)(3)(v)(B) and (D). Furthermore, examples 1 through 4 where LCOs were not entered and thus, multiple Completion Times were not met are Conditions Prohibited by TS and are reportable under 10 CFR 50.73(a)(2)(i)(B).

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences or radiological releases associated with these events.
The EDGs provide backup power if a loss of offsite power occurs. There are two fully redundant divisions, each having two EDGs. Therefore, there was no actual loss of safety function during the times when the Division 2 EDGs (13 and 14) were both OPERABLE while a Division 1 EDG (EDG 12) was inoperable.

The LOP instrumentation is implemented by two undervoltage functions, LOV relays and DVRs. Both the LOV relays and DVRs connected across phase-to-neutral of the Bus 64C PTs correctly responded to the blown PT fuses. The 130 V Direct Current (DC) supply for control of the undervoltage relays remained functional. The other relays (two for LOV and two for DVR scheme) connected across phase-to-phase of the Bus 64C PTs were not tripped. Therefore, the LOP instrumentation went from a one out of two, taken twice, logic to a one out of two logic and remained capable of fulfilling its design functions.

During a LOCA coincident with a degraded grid condition, the LOV relays and DVRs may not have reset starting at 1644 on April 24, 2016, until Operations increased the voltage above 121.3 meter V by manually using the SS64 LTC at 1920 on April 24, 2016. With the LOV relays and DVRs remaining in half-tripped position, the first occurrence of a degraded grid voltage may have led to premature separation of the Class 1E buses from the preferred offsite source, starting EDGs 11 and 12, and re-sequencing the safety loads. In this event, safety loads would have been powered by EDGs 11 and 12 to maintain the reactor in a safe condition. Although EDG 12 could not be manually synchronized with offsite power to allow loads to be transferred back to offsite power as a result of this condition, the EDGs have a seven day fuel supply, as required by TS 3.8.3. This would provide time for a temporary modification and procedure change to transfer the loads back to offsite power.

Therefore, although Division 1 EDG 12 should have been declared inoperable starting at 1644 on April 24, 2016, the inoperable LOP relays would not impact the capability of EDG 12 to start and assume loads during a design basis event. The LOP logic in the half tripped condition remained capable of starting the EDG upon a sensed loss of grid or degraded grid condition. Therefore, during the six timeframes when EDGs in both divisions were inoperable due to surveillances being performed on the Division 2 EDGs 13 and/or 14, both Division 1 EDGs remained capable of performing their safety-related functions to start and assume loads during a design basis event.

NARRATIVE

Furthermore, Division 2 offsite power remained functional throughout the event. There was no report of inoperability or malfunction with Class 1E equipment or onsite generation systems with Division 2. Therefore, reactor cooling and safety was always supported with at least one division of fully functional offsite power source and onsite generation.

CAUSE OF THE EVENT

The Direct Cause of the LOP instrumentation being inoperable was an open circuit. The phase-to-neutral PT high side fuses were blown on non-safety Bus 64A and safety-related Bus 64C, resulting in a loss of output from the PTs. A failure analysis performed by a third-party laboratory confirmed that the fuses had opened on a low energy transient event. The source of the low energy transient event could not be determined, but it appears to have been intermittent from either the 4160 V secondary side of SS64 or a transient on the 120 kV grid.

The cause of Operators not entering the required LCOs was unfamiliarity with this equipment condition and a failure to recognize the impacts of the condition.

CORRECTIVE ACTIONS

Transformer 64 Buses 64A, 64B, 64C, and 64V were inspected and the fuses were replaced during FO 16-01, which restored compliance with the TS requirements. A Temporary Modification was installed to monitor Bus 64A to detect future transients.

Operations completed a Department Event Free Day Reset and Event Review.

Other Corrective Actions include additional training for Operators on the issues described in this LER.

PREVIOUS OCCURRENCES

There are no previous LERs for similar events for a low energy transient event resulting in blown phase-to-neutral fuses on the PT relays.

LER 2015-009 documented an event on July 28, 2015, where Operators failed to enter an LCO during planned work on the UHS. In this event, the missed LCO entries were an inadvertent deletion by the Operators during preparation of the LCO log entries. Therefore, the Corrective Actions associated with LER 2015-009 would not have prevented the incorrect Operability determination and missed LCO entries discussed in this LER.

On January 6, 2016, an LCO entry was not made when the East and West Turbine Bypass Valves opened during High Pressure Stop Valve 1 drifted closed to 25%. This missed LCO entry was due to a non-conservative decision by the Operators. Contributing to this decision was a misinterpretation of an overly conservative statement in the TS Bases. This event was reported in LER 2016-001. Therefore, the Corrective Actions associated with this event would not have prevented the incorrect Operability determination and missed LCO entries discussed in this LER.

ADDITIONAL INFORMATION

Failed Component: Phase-to-neutral PT fuses
Manufacturer: General Electric
Model Number: EJ-1 1E